The z-Plane: Stability, Performance and the Jury Test

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1.0 Learning Outcomes

After studying this ACE-Lab learning guide, you should be able to demonstrate the Learning Outcomes listed in Table 1. On completion of the ACE-Lab learning guide, revisit Table 1 and confirm whether each Learning Outcome has been achieved.

Table 1: *Learning Outcomes from the z-Plane Stability and Performance Learning Guide*

ID	Learning Outcome	Achieved
1.	Explain how the s-plane maps onto the z-plane through $z = e^{sT_s}$, including the stable region, the unit circle and the lines of constant damping ratio and natural frequency.	☐
2.	Explain how the location of the poles on the z-plane, in Cartesian and polar form, determines the system response, and how the sampling interval affects the pole locations.	☐
3.	Determine the steady-state gain and the expected response of a discrete-time system from its transfer function, using the discrete final value theorem.	☐
4.	Apply the Jury stability test to determine whether a discrete-time closed-loop control system is stable, and the range of proportional control gain K_p that gives a stable system.	☐

2.0 Introduction

Understanding the dynamics of a closed-loop control system is important, and a model-based design approach allows changes to be explored before they are made on the real system. If potential issues are not identified, the result can be instability and failure of the system. Feedback control systems that have suffered from instability in practice include aircraft, helicopters with rotor imbalance, and wind turbines.

In the continuous-time Learning Guides, stability and performance were analysed on the s-plane, using the Routh-Hurwitz stability criterion and the root locus. Controllers are, however, almost always implemented on a digital microcontroller or computer, which samples the measured output at intervals of T_s seconds. The sampled-data control system in Figure 1 is described by a discrete-time transfer function $G(z)$, and its poles and zeros lie on the z-plane. An important question therefore arises:

Where must the poles of a discrete-time system lie for it to be stable, and how does their location set its performance?

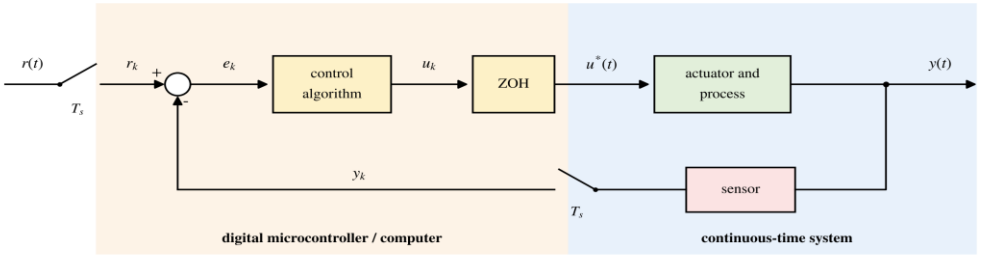


Figure 1: *Sampled-Data Closed-Loop Control System, with a Digital Controller and a Continuous-Time Actuator and Process*

The continuous-time actuator and process, together with the zero-order hold (ZOH) that holds each controller output constant over a sampling interval, are described by an equivalent discrete-time transfer function:

$$G(z) = \mathcal{Z}\{G_{ZOH}(s)\,G(s)\} \tag{1}$$

This gives the pure discrete-time equivalent in Figure 2. With a proportional controller K_p and an ideal sensor, the closed-loop transfer function has the same form as in continuous time:

$$G_{CL}(z) = \frac{Y(z)}{R(z)} = \frac{K_p G(z)}{1 + K_p G(z) H(z)} \tag{2}$$

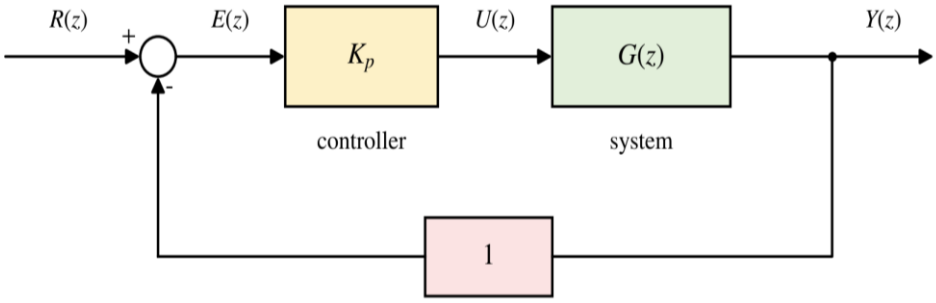


Figure 2: *Pure Discrete-Time Equivalent of the Sampled-Data Control System*

Why is Understanding the z-Plane Important?

The main reasons why the z-plane is important are:

i. The z-plane shows whether a digital control system is stable, which depends on the controller gain and on the sampling interval;
ii. The location of the poles on the z-plane sets the system response, just as on the s-plane, so performance can be designed directly in discrete time;
iii. The z-plane shows the effect of the sampling interval, which is a design choice in every digital controller;
iv. The z-plane is the basis of the Jury stability test and of root locus design in discrete time, covered in this and the next Learning Guide.

3.0 Mapping between the s-Plane and the z-Plane

The equivalent characteristics on the z-plane are related to those on the s-plane by:

$$z = e^{sT_s} \tag{3}$$

where T_s is the sampling interval, used throughout these Learning Guides. Writing $s = \sigma + j\omega$ gives $z = e^{\sigma T_s} e^{j\omega T_s}$, so the real part of s sets the distance of z from the origin, $|z| = e^{\sigma T_s}$, and the imaginary part sets its angle, ωT_s. This leads to the mapping shown in Figure 3 and summarised below:

- The entire left-half of the s-plane ($\sigma < 0$) maps onto the inside of the unit circle of the z-plane ($|z| < 1$), the stable region;
- The imaginary axis of the s-plane maps onto the unit circle ($|z| = 1$);
- The right-half of the s-plane maps onto the outside of the unit circle ($|z| > 1$), the unstable region;
- Real poles on the s-plane map onto real poles on the z-plane, and complex poles onto complex poles.

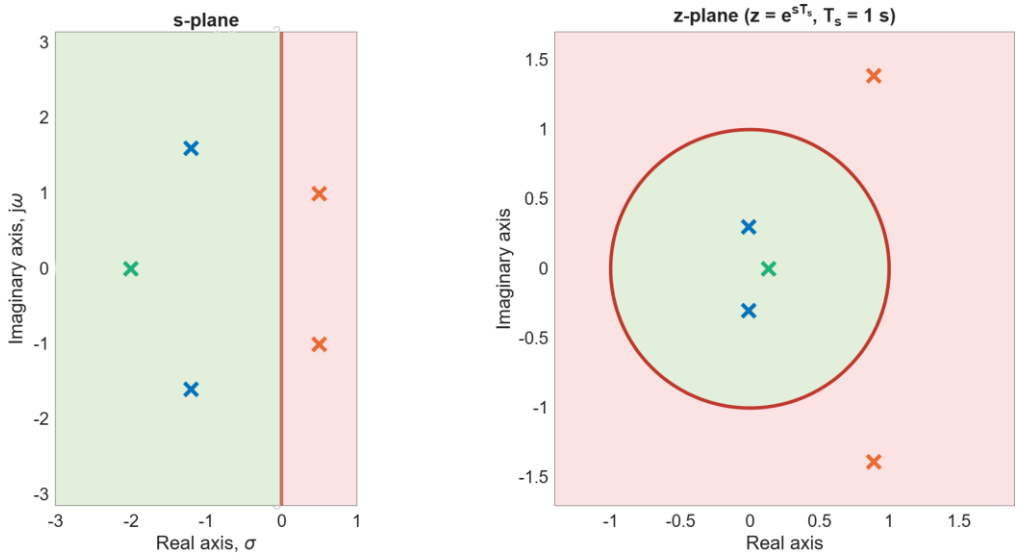


Figure 3: *Mapping from the s-Plane to the z-Plane with $T_s = 1$ s: the Stable Region Maps to the Inside of the Unit Circle*

Lines of constant natural frequency ω_n and damping ratio ζ intersect at right angles on the s-plane, and the same relationship holds on the z-plane. This is called conformal mapping. On the z-plane, lines of constant ζ become spirals from $z = 1$ towards the origin, and lines of constant ω_n become curves labelled in MATLAB as fractions of π/T_s, that is, $\omega_n = a\pi/T_s$. Both plots are symmetrical about the real axis.

Example 1: Natural frequency on the z-plane

A system has a natural frequency of $\omega_n = 2$ rad/s. On which constant-ω_n line of the MATLAB z-plane grid will its poles lie for sampling intervals of $T_s = 1$, 0.5 and 0.05 s?

Solution:

Setting $\omega_n = a\pi/T_s$ and solving for a:

$$a = \frac{\omega_n T_s}{\pi} = \frac{2(1)}{\pi} = 0.6366, \qquad \frac{2(0.5)}{\pi} = 0.3183, \qquad \frac{2(0.05)}{\pi} = 0.0318$$

The poles lie on the $0.64\pi/T_s$, $0.32\pi/T_s$ and $0.03\pi/T_s$ lines respectively. The smaller the sampling interval, the closer the poles are to $z = 1$.

4.0 Poles and Zeros on the z-Plane

Consider the continuous-time transfer function, with $\omega_n = 2$ rad/s and $\zeta = 0.6$:

$$G(s) = \frac{4}{s^2 + 2.4s + 4} \tag{4}$$

Discretised with a ZOH and a sampling interval of $T_s = 1$ s, this gives:

$$G(z) = \frac{0.7830z + 0.3253}{z^2 + 0.0176z + 0.0907} \tag{5}$$

The poles are found from the denominator using the quadratic formula, and the zero from the numerator:

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-0.0176 \pm \sqrt{0.0176^2 - 4(1)(0.0907)}}{2(1)}$$
$$= -0.0088 \pm j0.3011, \qquad z = -\frac{0.3253}{0.7830} = -0.4155 \tag{6}$$

Four to six decimal places are needed when mapping poles and zeros onto the z-plane, because small changes in the coefficients move the poles significantly. Figure 4 shows the poles on both planes, and Figure 5 the continuous-time and discrete-time step responses.

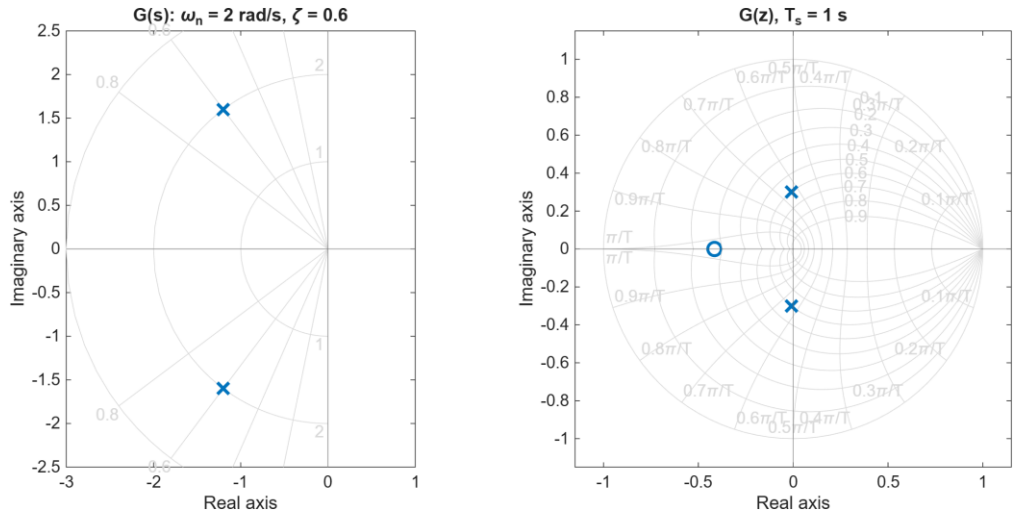


Figure 4: *Poles of $G(s)$ on the s-Plane, and Poles and Zero of $G(z)$ on the z-Plane, for $T_s = 1$ s*

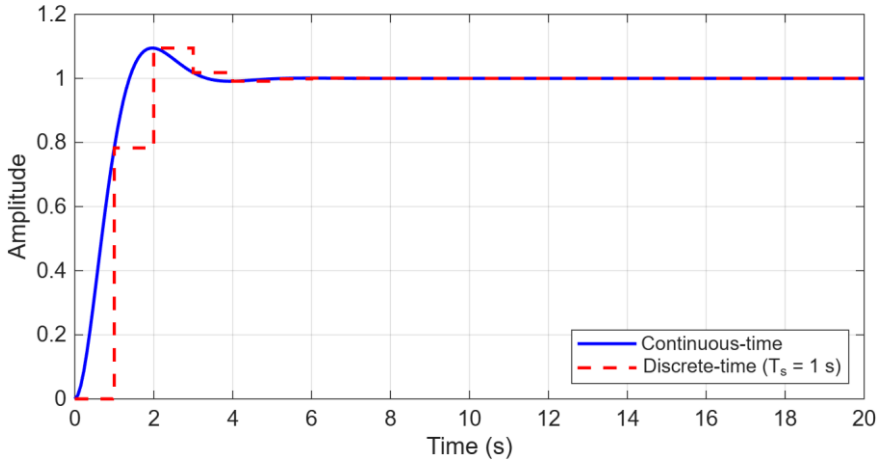


Figure 5: *Step Responses of $G(s)$ and of $G(z)$ with $T_s = 1$ s*

Pole Location and System Response

The location of the poles dictates the system response. A pole $z = x + jy$ can also be written in polar form, as shown in Figure 6:

$$z = x + jy = r\angle \pm \theta, \qquad (r, \theta) = \left(\sqrt{x^2 + y^2},\ \tan^{-1}\left|\frac{y}{x}\right|\right) \tag{7}$$

where the angle is measured from the positive real axis, taking account of the quadrant of the pole. From Equation (3), a continuous-time pole $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$ maps to:

$$r\angle \pm \theta = e^{-\zeta\omega_n T_s}\angle \pm \omega_n T_s\sqrt{1-\zeta^2} \tag{8}$$

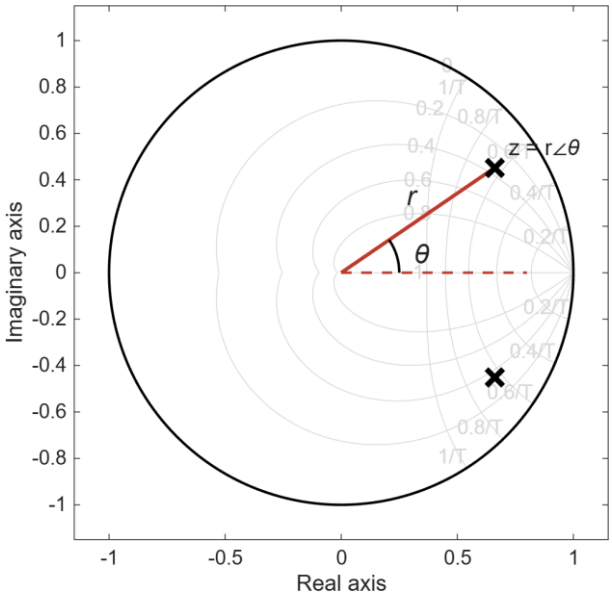


Figure 6: *Polar Form of a Pair of Complex Poles on the z-Plane*

The radius r therefore sets how quickly the response decays, and the angle θ sets how many samples each oscillation takes. In particular:

- $r < 1$ gives a decaying response;
- $r = 1$ gives a sustained (sinusoidal) response;
- $r > 1$ gives an increasing response.

A pole on the positive real axis gives a response that does not oscillate, while a pole on the negative real axis gives a response that alternates in sign at every sample, a behaviour with no continuous-time equivalent. These cases are shown in Figure 7. Rearranging Equation (8) gives the damping ratio and natural frequency of a pair of complex poles:

$$\zeta\omega_n = -\frac{\ln r}{T_s}, \qquad \omega_d = \omega_n\sqrt{1-\zeta^2} = \frac{\theta}{T_s} \tag{9}$$

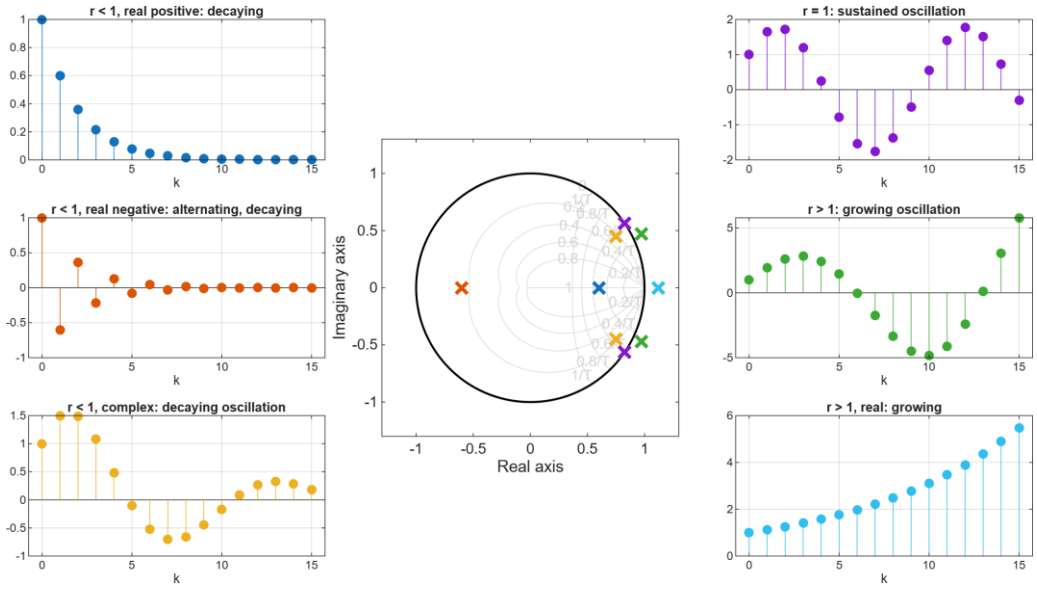


Figure 7: *Pole Locations on the z-Plane and the Corresponding Impulse Responses*

Effect of the Sampling Interval

The sampling interval changes the discrete-time transfer function and moves the poles. Consider $G(s) = \frac{12}{s^2+5s+6}$, which has real poles at $s = -2$ and -3. Discretising with three sampling intervals gives:

Table 2: *Discrete-Time Transfer Function and Poles of* $G(s) = 12/(s^2 + 5s + 6)$ *for Three Sampling Intervals*

T_s **(s)**	$G(z)$	**Poles,** e^{-2T_s} **and** e^{-3T_s}
1	$\frac{1.3871z + 0.2561}{z^2 - 0.1851z + 0.0067}$	0.1353, 0.0498
0.5	$\frac{0.6852z + 0.2969}{z^2 - 0.5910z + 0.0821}$	0.3679, 0.2231
0.05	$\frac{0.0138z + 0.0127}{z^2 - 1.7655z + 0.7788}$	0.9048, 0.8607

As shown in Figure 8, a smaller sampling interval moves the poles towards $z = 1$, and the discrete-time response approaches the continuous-time response. Because the poles then crowd together near $z = 1$, more decimal places are needed to describe them accurately. The sampling interval must also be small enough to capture the dynamics: as a guide, the angle $\theta = \omega_d T_s$ should be well below π, or the oscillation is not represented correctly.

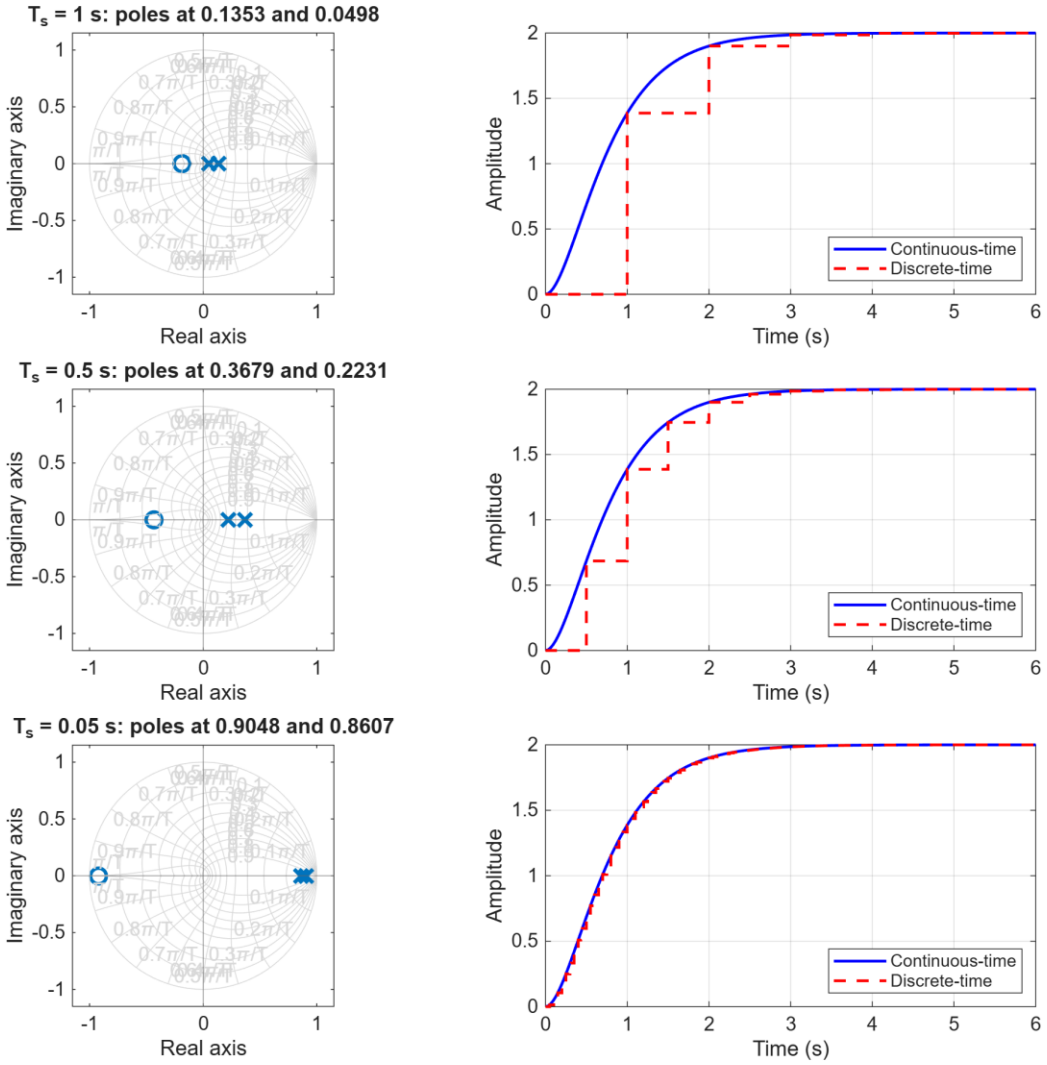


Figure 8: *Pole Locations and Step Responses of* $G(s) = 12/(s^2 + 5s + 6)$ *for* $T_s = 1,\ 0.5$ *and* 0.05 *s*

5.0 System Properties and Response

System Performance Criteria

The system performance criteria (SPC) are the same in discrete time as in continuous time. For an underdamped system subject to a step input, the following are important, as shown in Figure 9:

i. rise time, t_r;
ii. peak time, t_p, and period of oscillation, T;
iii. peak overshoot amplitude, A_p;
iv. settling time, t_s, within a given percentage of the final value (for example 5%);
v. number of oscillations, N, within the settling time.

In discrete time, each of these is measured in samples and converted to seconds by multiplying by T_s. MATLAB can be used to determine the SPC, for example with the stepinfo command.

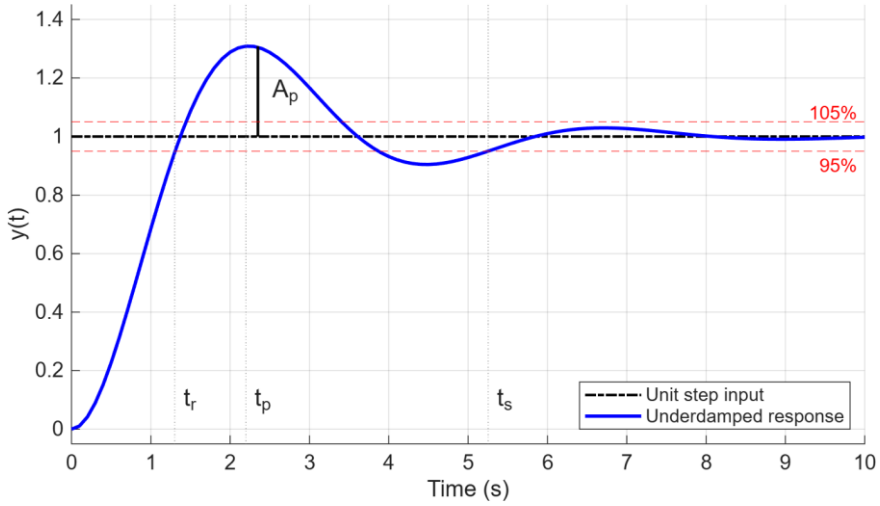


Figure 9: *System Performance Criteria for an Underdamped Step Response*

The Discrete Final Value Theorem

The final value theorem gives the final value of a system response when it reaches steady state, without the response having to be simulated. It can be applied to an open-loop system or a closed-loop control system, and for a closed-loop control system it gives the steady-state error (SSE). The discrete form of the final value theorem is:

$$\lim_{n\to\infty} f[n] = \lim_{z\to 1}(z-1)F(z) \tag{10}$$

where $n = 1,2,3,\ldots$ is the sample number, and the samples are taken at times $t_0, t_1, t_2, \ldots$. The theorem only applies if the system is stable. For a unit step input, $R(z) = z/(z-1)$, so the final value of the output is simply $G(1)$, the **steady-state gain** (SSG).

Example 2: Steady-state gain, poles and expected response

Determine the steady-state gain (SSG) of the following transfer function, which is the final value of the system response to a unit step input:

$$G(z) = \frac{0.0036z + 0.0035}{z^2 - 1.8930z + 0.9048}$$

Based on the discrete transfer function, calculate and sketch: (i) the poles and zeros on the z-plane; (ii) the polar form of the poles; (iii) the expected system response to a unit step input, including the transient and steady state.

Solution:

SSG: setting $z = 1$:

$$G(1) = \frac{0.0036 + 0.0035}{1 - 1.8930 + 0.9048} = \frac{0.0071}{0.0118} = 0.6017$$

(i) The poles are:

$$z = \frac{1.8930 \pm \sqrt{1.8930^2 - 4(0.9048)}}{2} = 0.9465 \pm j0.0945$$

and the zero is at $z = -0.0035/0.0036 = -0.9722$.

(ii) In polar form:

$$r = \sqrt{0.9465^2 + 0.0945^2} = 0.9512, \qquad \theta = \tan^{-1}\left(\frac{0.0945}{0.9465}\right) = 0.0996 \text{ rad } (5.7°)$$

so, the poles are $z = 0.9512\angle \pm 5.7°$. Note that $r^2 = 0.9048$, the constant term of the denominator.

(iii) Since $r < 1$ the response decays, and since θ is small the oscillation is slow, about $2\pi/0.0996 = 63$ samples per cycle. Because r is close to 1, the decay is also slow. The response is therefore an underdamped rise to a final value of 0.60, as shown in Figure 10. Using Equation (9), the poles correspond to $\zeta = 0.45$ and $\omega_n T_s = 0.111$; the peak overshoot is approximately 21%, after 32 samples.

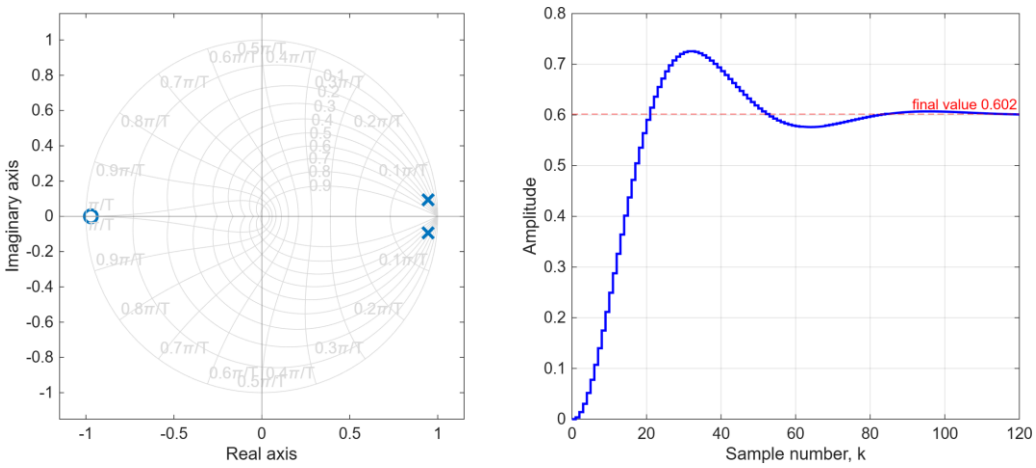


Figure 10: *Pole-Zero Map and Unit Step Response for Example 2*

MATLAB Script 1: check of Example 2

```
clear; clc; close all;
num = [0.0036 0.0035];
den = [1 -1.8930 0.9048];
G = tf(num,den,-1)    % discrete, unspecified sampling interval
SSG = dcgain(G)
p = pole(G)
r = abs(p), theta = angle(p)
figure; pzmap(G); zgrid
figure; step(G)
```

Output:

```
SSG = 0.6017
p = 0.9465 + 0.0945i
    0.9465 - 0.0945i
r = 0.9512
```

```
theta = 0.0996
```

6.0 The Jury Stability Test

A proportional control gain K_p that is too large moves the closed-loop poles outside the unit circle, making the system unstable. The Jury test is an algebraic test, similar in form to the Routh-Hurwitz stability criterion, that determines whether all the roots of a polynomial lie inside the unit circle, without calculating them.

Necessary and Sufficient Conditions

The characteristic equation of the closed-loop control system is written in the general form:

$$P(z) = a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n = 0, \qquad a_0 > 0 \tag{11}$$

The following conditions are necessary and sufficient for a stable system, that is, for all roots to lie inside the unit circle:

i. $|a_n| < a_0$;
ii. $P(1) > 0$;
iii. $P(-1) > 0$ for n even, or $P(-1) < 0$ for n odd;
iv. $|b_{n-1}| > |b_0|$, $|c_{n-2}| > |c_0|$, and so on, for each pair of rows of the Jury table.

The first three conditions are quick to check and should be tested first: if any fails, the system is unstable and the table is not needed. For a second-order system ($n = 2$), the first three conditions are sufficient.

Populating the Jury Table

The Jury table is built from the coefficients of the characteristic equation, as shown in Table 3. It has $2n - 3$ rows. Row 1 contains the coefficients in reverse order and row 2 in forward order. Each later pair of rows is calculated from the pair above, and the second row of each pair is the first row reversed.

Table 3: *General Form of the Jury Table*

Row	z^0	z^1	z^2	…	z^{n-1}	z^n
1	a_n	a_{n-1}	a_{n-2}	$\cdots$	a_1	a_0
2	a_0	a_1	a_2	$\cdots$	a_{n-1}	a_n
3	b_{n-1}	b_{n-2}	b_{n-3}	$\cdots$	b_0	
4	b_0	b_1	b_2	$\cdots$	b_{n-1}	
5	c_{n-2}	c_{n-3}	c_{n-4}	$\cdots$		
6	c_0	c_1	c_2	$\cdots$		

Row	z^0	z^1	z^2	⋯	z^{n-1}	z^n
⋮	⋮	⋮	⋮			

The elements of rows 3 and 5 are calculated from the determinants:

$$b_k = \begin{vmatrix} a_n & a_{n-1-k} \\ a_0 & a_{k+1} \end{vmatrix}, \qquad k = 0,1,2,\dots,n-1 \tag{12}$$

$$c_k = \begin{vmatrix} b_{n-1} & b_{n-2-k} \\ b_0 & b_{k+1} \end{vmatrix}, \qquad k = 0,1,2,\dots,n-2 \tag{13}$$

where $\begin{vmatrix} p & q \\ r & s \end{vmatrix} = ps - qr$. The process continues until the last row contains three elements.

Example 3: Stability of a fourth-order system

The characteristic equation of a feedback control system is:

$$P(z) = z^4 - 1.2z^3 + 0.07z^2 + 0.3z - 0.08 = 0$$

Determine whether the system is stable, using the Jury test.

Solution:

The coefficients are $a_0 = 1$, $a_1 = -1.2$, $a_2 = 0.07$, $a_3 = 0.3$ and $a_4 = -0.08$, with $n = 4$, so the table has $2n - 3 = 5$ rows. The first three conditions are:

$$|a_4| = 0.08 < a_0 = 1, \qquad P(1) = 1 - 1.2 + 0.07 + 0.3 - 0.08 = 0.09 > 0$$

$$P(-1) = 1 + 1.2 + 0.07 - 0.3 - 0.08 = 1.89 > 0 \quad (n \text{ even})$$

All three are satisfied. Row 3 is calculated from Equation (12), for example:

$$b_3 = \begin{vmatrix} -0.08 & 1 \\ 1 & -0.08 \end{vmatrix} = 0.0064 - 1 = -0.9936, \qquad b_0 = \begin{vmatrix} -0.08 & 0.3 \\ 1 & -1.2 \end{vmatrix} = 0.096 - 0.3 = -0.2040$$

giving $b_1 = -0.0756$ and $b_2 = 1.1760$. Row 5 is calculated from Equation (13):

$$c_2 = \begin{vmatrix} -0.9936 & -0.2040 \\ -0.2040 & -0.9936 \end{vmatrix} = 0.9456, \qquad c_0 = \begin{vmatrix} -0.9936 & 1.1760 \\ -0.2040 & -0.0756 \end{vmatrix} = 0.3150$$

and $c_1 = -1.1839$. The completed table is:

Row	z^0	z^1	z^2	z^3	z^4
1	-0.08	0.3	0.07	-1.2	1
2	1	-1.2	0.07	0.3	-0.08
3	-0.9936	1.1760	-0.0756	-0.2040	
4	-0.2040	-0.0756	1.1760	-0.9936	
5	0.9456	-1.1839	0.3150		

The fourth condition is now checked:

$$|b_3| = 0.9936 > |b_0| = 0.2040, \qquad |c_2| = 0.9456 > |c_0| = 0.3150$$

All the conditions are satisfied, so the feedback control system is **stable**. (Its roots are 0.8, 0.5, 0.4 and −0.5, all inside the unit circle.)

Example 4: Range of proportional control gain for stability

The characteristic equation of a feedback control system is:

$$P(z) = z^3 - 1.5z^2 + \big(K_p + 1\big)z - 0.25 = 0$$

where $K_p > 0$ is the controller gain. Determine the range of K_p for which the system is stable.

Solution:

The coefficients are $a_0 = 1$, $a_1 = -1.5$, $a_2 = K_p + 1$ and $a_3 = -0.25$, with $n = 3$, so the table has $2n - 3 = 3$ rows. The first three conditions are:

$$|a_3| = 0.25 < a_0 = 1, \qquad P(1) = 1 - 1.5 + \big(K_p + 1\big) - 0.25 = 0.25 + K_p > 0$$

$$P(-1) = -1 - 1.5 - \big(K_p + 1\big) - 0.25 = -3.75 - K_p < 0 \;\; (n \text{ odd})$$

All three are satisfied for every $K_p > 0$. Row 3 is calculated from Equation (12):

$$b_2 = \begin{vmatrix} -0.25 & 1 \\ 1 & -0.25 \end{vmatrix} = 0.0625 - 1 = -0.9375$$

$$b_1 = \begin{vmatrix} -0.25 & -1.5 \\ 1 & K_p + 1 \end{vmatrix} = 1.25 - 0.25K_p, \qquad b_0 = \begin{vmatrix} -0.25 & K_p + 1 \\ 1 & -1.5 \end{vmatrix} = -0.625 - K_p$$

The fourth condition requires:

$$|b_2| > |b_0| \;\Rightarrow\; 0.9375 > 0.625 + K_p \;\Rightarrow\; K_p < 0.3125$$

All the criteria are therefore satisfied for $0 < K_p < 0.3125$. At $K_p = 0.3125$, a pair of roots lies on the unit circle.

MATLAB Script 2: check of Examples 3 and 4 using the magnitude of the roots

```
clear; clc;
%% Example 3
P = [1 -1.2 0.07 0.3 -0.08];
abs(roots(P))'          % all less than 1: stable
%% Example 4: largest root magnitude as Kp varies
Kp = 0:0.005:0.5;
rmax = zeros(size(Kp));
for i = 1:length(Kp)
    rmax(i) = max(abs(roots([1 -1.5 Kp(i)+1 -0.25])));
end
plot(Kp,rmax); grid on
xlabel('K_p'); ylabel('Largest |z|')
```

Output:

```
ans =
    0.8000    0.5000    0.5000    0.4000
```

For Example 4, the largest root magnitude reaches 1 at $K_p = 0.3125$, confirming the Jury result.

7.0 Exercises

Attempt the following exercises. MATLAB may be used to check the answers.

a) A continuous-time system is modelled by $G(s) = \frac{12}{s^2+s+12}$. Using MATLAB, discretise it with $T_s = 1$ s to determine $G(z)$. Determine the steady-state gain, and calculate and sketch the poles and zeros, the polar form of the poles and the expected response to a unit step input. Comment on the choice of sampling interval.

b) Map the continuous-time poles $s = -1 \pm j2$ onto the z-plane for $T_s = 0.1$ s, giving each in Cartesian and polar form.

c) Use the Jury test to determine whether the system with characteristic equation $P(z) = z^2 - 0.7z + 0.1 = 0$ is stable.

d) Use the Jury test to determine whether the system with characteristic equation $P(z) = z^3 - 1.3z^2 - 0.08z + 0.24 = 0$ is stable.

e) A closed-loop control system has the characteristic equation $P(z) = z^3 - 1.2z^2 + 0.36z + 0.1K_p = 0$. Use the Jury test to determine the range of K_p for which it is stable.

f) The system $G(z) = \frac{1.3872z+0.2561}{(z-0.1353)(z-0.0498)}$ is placed in a closed loop with a proportional control gain K_p and negative unity feedback. Determine the characteristic equation and the range of K_p that gives a stable system.

Fully worked solutions to the above exercises can be found in **Appendix A**.

8.0 Summary

This Learning Guide extended stability and performance analysis to discrete-time systems. The mapping $z = e^{sT_s}$ takes the left-half of the s-plane to the inside of the unit circle, so a discrete-time system is stable when all of its poles satisfy $|z| < 1$. Lines of constant damping ratio and natural frequency carry across to the z-plane by conformal mapping.

In polar form, $z = r\angle\theta$, the radius sets the decay ($r < 1$ decays, $r = 1$ is sustained, $r > 1$ grows) and the angle sets the oscillation. A smaller sampling interval moves the poles towards $z = 1$ and makes the discrete-time response closer to the continuous-time one. The discrete final value theorem gives the steady-state gain as $G(1)$.

The Jury test determines stability algebraically, from three simple conditions and a table of determinants, and with K_p kept as a symbol it gives the range of gain for a stable system. The next Learning Guide uses these results to design a proportional controller with the root locus on the z-plane.

Appendix A: Fully Worked Solutions to Exercises

Question a:

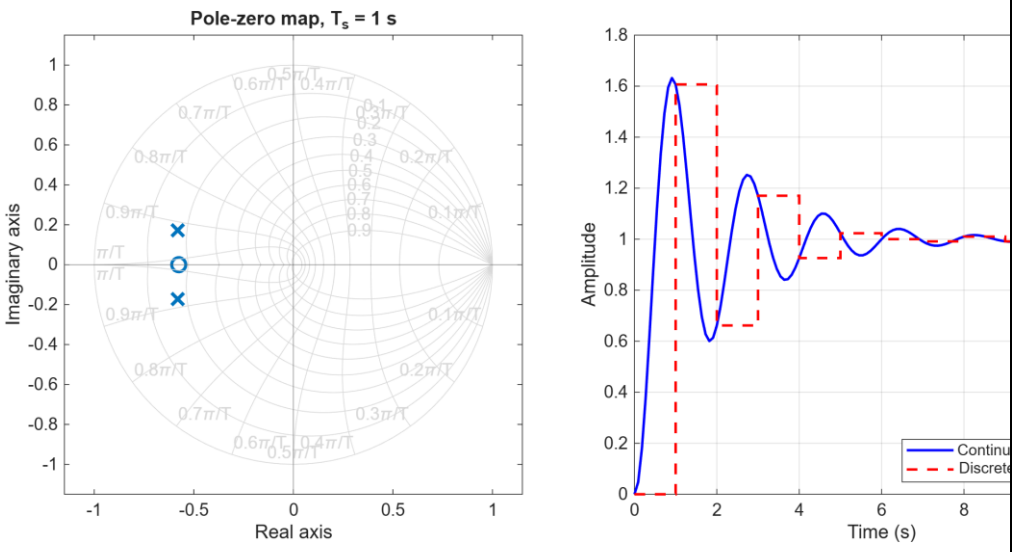
```
G = tf(12,[1 1 12]);
Gz = c2d(G,1,'zoh')
```

$$G(z) = \frac{1.6068z + 0.9248}{z^2 + 1.1637z + 0.3679}$$

SSG: $G(1) = (1.6068 + 0.9248)/(1 + 1.1637 + 0.3679) = 2.5316/2.5316 = 1$, as for the continuous-time system.

Poles and zero: $z = -0.5819 \pm j0.1712$ and $z = -0.5755$. **Polar form:** $r = 0.6065$ and $\theta = 163.6°$ (2.855 rad).

Since $r < 1$ the response decays quickly, but the poles lie close to the negative real axis, so the sampled response alternates from sample to sample, as shown below. The continuous-time poles have $\omega_d = \sqrt{11.75} = 3.43$ rad/s, so $\omega_d T_s = 3.43$ rad, which is greater than π: the sampling interval is too long to represent the oscillation, and the angle on the z-plane is folded back (aliased). A much shorter interval should be used; with $T_s = 0.1$ s the poles are $0.9512\angle \pm 19.6°$ and the sampled response follows the continuous one.



Question b:

$$z = e^{(-1 \pm j2)(0.1)} = e^{-0.1}\angle \pm 0.2 \text{ rad} = 0.9048\angle \pm 11.46°$$

$$z = 0.9048(\cos 0.2 \pm j \sin 0.2) = 0.8868 \pm j0.1798$$

Question c:

With $n = 2$, the first three conditions are sufficient:

$$|a_2| = 0.1 < 1, \qquad P(1) = 1 - 0.7 + 0.1 = 0.4 > 0, \qquad P(-1) = 1 + 0.7 + 0.1 = 1.8 > 0$$

All are satisfied, so the system is **stable** (roots 0.5 and 0.2).

Question d:

$$|a_3| = 0.24 < 1, \qquad P(1) = 1 - 1.3 - 0.08 + 0.24 = -0.14 < 0$$

The second condition fails, so the system is **unstable** and the table is not needed. (One root is at $z = 1.2$, outside the unit circle.)

Question e:

With $a_0 = 1$, $a_1 = -1.2$, $a_2 = 0.36$ and $a_3 = 0.1K_p$:

$$|a_3| < 1 \Rightarrow K_p < 10, \qquad P(1) = 0.16 + 0.1K_p > 0, \qquad P(-1) = -2.56 + 0.1K_p < 0 \Rightarrow K_p < 25.6$$

$$b_2 = \left(0.1K_p\right)^2 - 1, \qquad b_0 = \begin{vmatrix} 0.1K_p & 0.36 \\ 1 & -1.2 \end{vmatrix} = -0.12K_p - 0.36$$

For $0 < K_p < 10$, $|b_2| = 1 - 0.01K_p^2$ and $|b_0| = 0.36 + 0.12K_p$, so the fourth condition requires:

$$1 - 0.01K_p^2 > 0.36 + 0.12K_p \;\Rightarrow\; K_p^2 + 12K_p - 64 < 0 \;\Rightarrow\; \left(K_p + 16\right)\left(K_p - 4\right) < 0$$

The system is therefore stable for $0 < K_p < 4$.

Question f:

The characteristic equation is $(z - 0.1353)(z - 0.0498) + K_p(1.3872z + 0.2561) = 0$, that is:

$$P(z) = z^2 + \left(1.3872K_p - 0.1851\right)z + \left(0.0067 + 0.2561K_p\right) = 0$$

$$|a_2| < 1 \Rightarrow K_p < 3.88, \qquad P(1) = 0.8216 + 1.6433K_p > 0$$

$$P(-1) = 1.1918 - 1.1311K_p > 0 \;\Rightarrow\; K_p < 1.054$$

The system is stable for $0 < K_p < 1.054$. At this gain a closed-loop pole reaches $z = -1$. This system is the starting point of the next Learning Guide.

Recommended Citation

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